



Course : R-VIJETA (R-JP)

PHYSICS DPP

DAILY PRACTICE PROBLEM

REVISION DPP NO. # 07

Target : JEE(Advanced) 2023

DPP Syllabus :

KTG, HEAT, THERMODYNAMICS, CALORIMETRY, THERMAL EXPANSION & HEAT TRANSFER.

DPP No. : 07

Maximum Marks : 170

Total Time : 160 Min.

One or more than one may be correct option ('-1' Negative marking) Q.1 to 18;

(3 Marks; 2 Min.) [54, 36]

Comprehension ('-1' Negative marking) Q. 19 to Q. 20 ;

(3 Marks; 2 Min.) [06, 04]

Subjective Questions ('-1' Negative marking) Q. 21 to Q. 40 ;

(4 Marks; 5 Min.) [80, 100]

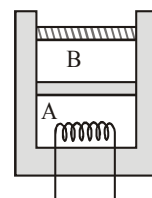
Single Choice Objective ('-1' Negative marking) Q. 41;

(3 Marks; 2 Min.) [03, 02]

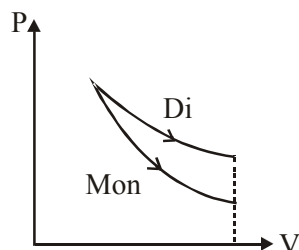
Comprehension ('-1' Negative marking) Q. 42 to Q. 50 ;

(3 Marks; 2 Min.) [27, 18]

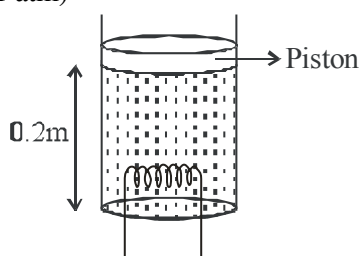
- The temperature of equal masses of three different liquids A, B and C is 12, 18 and 28°C, respectively. When A and B are mixed the temperature is 16°C. When B and C are mixed, it is 23°C. Select correct statements
 (A) The ratio of specific heats of liquid A and B is 1 : 2
 (B) The ratio of specific heats of liquid B and C is 1 : 3
 (C) The ratio of specific heats of liquid A and C is 1 : 2
 (D) When A and C are mixed, the equilibrium temperature will be 20°C
- Process undergone by an ideal gas sample is described in terms of thermodynamic temperature (T) and volume occupied by gas (V) by the equation $TV^x = \text{constant}$. Degrees of freedom of the gas sample is f. Choose the correct statements.
 (A) For $fx = 2$, the process represents adiabatic.
 (B) For $fx = 1$, temperature of the gas sample should decrease if heat is supplied to the gas.
 (C) For $x = -1$, coefficient of cubical expansion of the gas is equal to reciprocal of thermodynamic temperature.
 (D) For $x = -1$, bulk modulus of the gas sample is zero.
- Cylinder has 2 pistons each can move without friction. Bottom piston is conducting but the above piston is perfectly insulating. Lower chamber (A) carries 2 moles of monoatomic gas and upper chamber (B) carries 4 moles diatomic gas. Through a heater system is slowly heated and a total heat of 190R is given:
 (A) The increase in temperature of both the gasses is 10 K.
 (B) Work done by gas in chamber B is 40 R joules
 (C) Work done by gas in chamber A is 40 R joules
 (D) Both the gasses will suffer equal change in volume.



4. The ends of a rod of uniform thermal conductivity are maintained at different (constant) temperatures. After the steady state is achieved :
- heat flows in the rod from high temperature to low temperature even if the rod has nonuniform cross sectional area.
 - temperature gradient along length is same even if the rod has non uniform cross sectional area.
 - heat current is same even if the rod has non-uniform cross sectional area.
 - if the rod has uniform cross sectional area the temperature is same at all points of the rod.
5. Two Ideal gas samples (one monatomic & other diatomic) can be expanded from same initial state to a certain volume through adiabatic process. Choose the correct options.

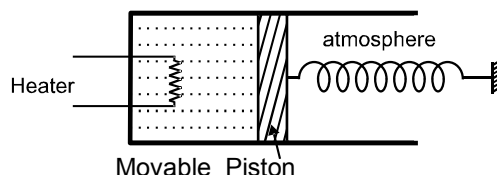


- Final temperature in monatomic gas will be greater than diatomic gas
 - Final temperature in Diatomic gas will be greater than monatomic gas
 - Final pressure in monatomic gas will be greater than in diatomic gas
 - Final pressure in diatomic gas will be greater than in monatomic gas.
6. Consider an insulated cylinder containing nitrogen gas and sealed on top by a heavy non conducting piston, as shown in figure. The piston is free to slide up and down the cylinder, with negligible friction. Initially, the piston is 0.20 m above the bottom of the cylinder, and the gas is at room temperature, 300 K. But then a student provides heat to the gas through a heater. As a result the piston slowly rises until reaching a height 0.30 m above the bottom of the cylinder, at which point the student extinguish the flame. (atmospheric pressure is 1 atm)



- Immediately before the flame gets ignited, the pressure of nitrogen inside the cylinder is greater than 1 atm
- Immediately before the flame gets ignited, the pressure of nitrogen inside the cylinder is equal to 1 atm
- When piston reaches a height of 0.30 m, the temperature of the nitrogen inside the cylinder is most nearly 450 K
- Work done by the nitrogen on the piston as it arises from 0.20m to 0.30 m is W. During that time interval the net heat absorbed by the nitrogen is equal to W

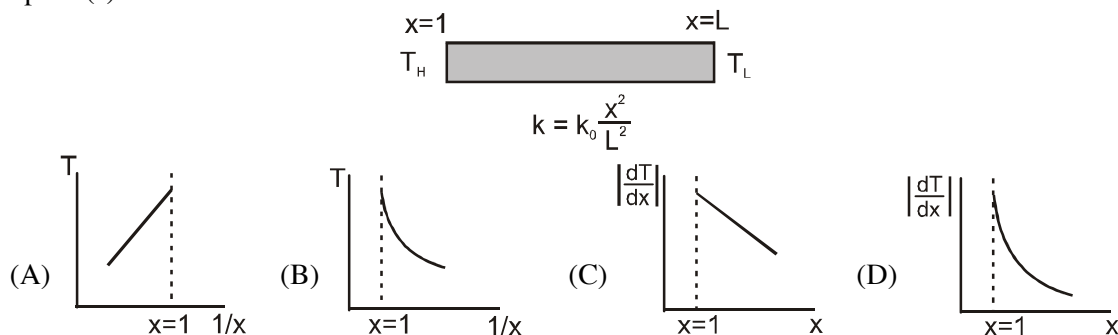
7. In an isobaric process (γ is adiabatic exponent of the gas) :
- (A) The heat given to gas is $\frac{\gamma}{\gamma-1}$ times the work done by gas.
 (B) The work done by gas is $(\gamma-1)$ times the change in internal energy.
 (C) The temperature of gas is increased.
 (D) The temperature of gas is decreased.
8. When the temperature of a copper coin is raised by 80°C , its diameter increases by 0.2% . Then choose the correct option(s) :
- (A) percentage rise in the area of a face is 0.4%
 (B) percentage rise in the thickness is 0.4%
 (C) percentage rise in the volume is 0.6%
 (D) coefficient of linear expansion of copper is $0.25 \times 10^{-4}/^\circ\text{C}$
9. An ideal gas is filled in a fixed adiabatic cylinder as shown in figure. The initial temperature, pressure and volume of the gas are T_0 , P_0 and V_0 respectively where P_0 is atmospheric pressure. A light non conducting and smooth piston of area A is connected to a spring of spring constant K , which is initially in natural length. Now the gas is heated slowly for some time, due to which the piston moves out slowly by a distance ' x '. Then :



- (A) Pressure of the gas is $P_0 + \frac{Kx}{A}$
 (B) Temperature of the gas is $\left(1 + \frac{Kx}{P_0 A}\right) \left(1 + \frac{Ax}{V_0}\right) T_0$
 (C) The gas is undergoing constant pressure process
 (D) Work done by the gas is $\frac{1}{2} Kx^2$
10. One mole of an ideal monoatomic gas undergoes a cyclic process 'abca' shown in figure. If U denotes the internal energy and W the work done by the gas. Then choose the **incorrect** alternative (s)
- (A) $U_c - U_a = 10.5 RT_0$
 (B) $U_b - U_a = +4RT_0$
 (C) $W_{c \rightarrow a} = -3P_0 V_0$
 (D) $U_c > U_b > U_a$
-
- The P-V diagram shows a cyclic process 'abca'. The vertical axis is Pressure (P) and the horizontal axis is Volume (V). The process starts at point 'a' with pressure P_0 and volume V_0 . It goes to point 'b' with pressure P_0 and volume $4V_0$ (isobaric expansion). From 'b', it goes to point 'c' with pressure $2P_0$ and volume $4V_0$ (isochoric heating). From 'c', it goes back to point 'a' with pressure P_0 and volume V_0 (adiabatic compression). The temperature at 'a' is T_0 .
11. A metal cylinder of 5 kg is heated electrically by a 20 W heater in a room at 20°C . The cylinder temperature rises to 30°C in 5 minutes and finally becomes constant at 45°C . Assuming that the rate of heat loss is proportional to the excess temperature over the surroundings. Choose the correct options :
- (A) The rate of loss of heat of cylinder to surrounding at 25°C is 4 W
 (B) The rate of loss of heat of cylinder to surrounding at 45°C is 20 W
 (C) The rate of loss of heat of cylinder to surrounding at 25°C is 8 W
 (D) The rate of loss of heat of cylinder to surrounding at 45°C is 0 W

12. The emissive power of a black body at $T = 300 \text{ K}$ is 100 Watt/m^2 . Consider a body B of area $A = 10 \text{ m}^2$ reflecting power $r = 0.3$ and transmission power $t = 0.5$. Its temperature is 300 K . Then which of the following is/are correct :
- (A) The emissive power of B is 20 W/m^2 (B) The emissive power of B is 200 W/m^2
 (C) The power emitted by B is 200 Watts (D) The emissivity of B is 0.2

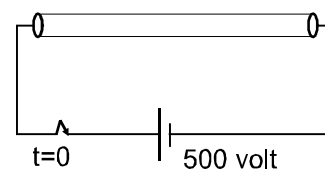
13. A rod of uniform cross-section but non-uniform thermal conductivity which vary as $k = k_0 \frac{x^2}{L^2}$ ($1 \leq x \leq L$) (as shown in figure) is kept between fixed temperature difference for a long time. Select the correct option(s) :



14. A straight nicrome wire is initially at room temperature 20°C . It is connected to an ideal battery of 500 volt . Just after switching on, the current detected is 5 amp . Due to heating effect its temperature increases, and is also losing heat to the environment according to Newton's law of cooling as $\frac{dQ_{\text{loss}}}{dt} = 45$

($T - 20^\circ\text{C}$) J/sec. At steady state, the current detected is 4.5 amp .

- (A) steady state temperature of the wire is 70°C
 (B) steady state temperature of the wire is 75.5°C
 (C) temperature co-efficient of resistance of the wire is nearly $2.2 \times 10^{-3} / ^\circ\text{C}$
 (D) temperature co-efficient of resistance of the wire is nearly $1.57 \times 10^{-3} / ^\circ\text{C}$



15. Expression for Maxwell's velocity distribution curve is given as $\frac{dN}{dV} = aNV^2 e^{-bV^2}$ for an ideal gas. Here, N is total number of molecule and a & b are positive constant. Which of the following is/are correct? Here, R , T and M are universal gas constant, absolute temperature and molar mass of the gas:

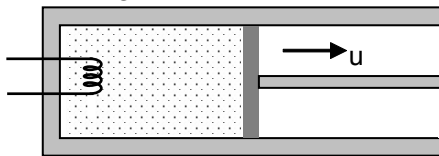
- (A) $b = \frac{M}{2RT}$ (B) $b = \frac{M}{3RT}$ (C) $a = \sqrt{\frac{2M^3}{\pi R^3 T^3}}$ (D) $a = \left[\frac{M}{2\pi RT} \right]^{3/2} \times 8\pi$

16. Water at 50°C is filled in a cubical container of side 1 m . The thickness of the walls of the container is 1 mm . The container is surrounded by large amount of ice at 0°C . The temperature of the water becomes 25°C in $10 \ln 2$ seconds. Choose the correct option(s)

[Given, specific heat of water = $1 \text{ cal/gm } ^\circ\text{C}$, latent heat of fusion of ice = $80 \text{ cal/gm } ^\circ\text{C}$, density of water = 1 gm/cm^3 ; heat capacity of the container ≈ 0]

- (A) thermal conductivity of the material is $70 \text{ J/m } ^\circ\text{C}$
 (B) thermal conductivity of the material is $60 \text{ J/m } ^\circ\text{C}$
 (C) Mass of the ice melted is 312.5 kg
 (D) Mass of the ice melted is 252 kg

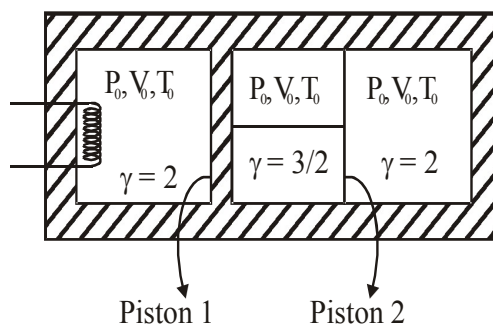
17. Ideal gas at pressure ' p_0 ' and volume ' V_0 ' is enclosed by a piston in a cylinder. We start to move the piston outwards at a constant velocity ' u '. The cross-sectional area of the piston is A . While the piston is moving, we can deliver heat to the gas through a heating filament so that temperature of gas remains constant. Suppose the power of heating filament is P . Select correct statements.



- (A) Pressure of the gas at any instant t is $\frac{p_0 V_0}{V_0 + A u t}$
- (B) Power should be $p_0 A u$ at beginning.
- (C) Power of filament should remain constant.
- (D) Power of filament will decrease with time.
18. An ideal monatomic gas expands in such a way that $TV^{1/2} = \text{constant}$, where T and V are the temperature & volume of the gas. Choose the correct option(s) :
- (A) The molar heat capacity is $C = C_V - 2R$. (B) The internal energy of gas decreases.
- (C) Temperature of gas decreases. (D) Pressure of gas increases.

Paragraph for Questions Nos. 19 to 20

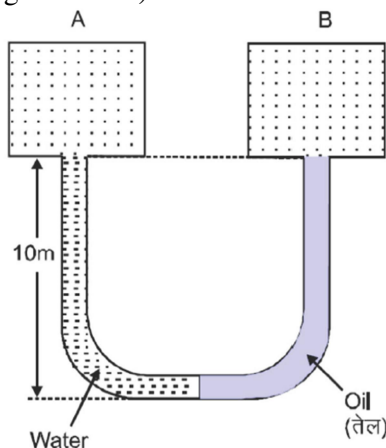
We have an adiabatic chamber as shown, which is initially divided in three parts having equal volume. Each part contains gases at same pressure and temperature. Piston - 1 is adiabatic and piston 2 is conducting. Piston - 1 and piston 2 are connected by a non - conducting rigid rod, which has negligible heat capacity and negligible temperature coefficient of linear expansion. Adiabatic exponent of gas filled in left most and right most part of the container is 2 and that of gas filled in middle part is $\frac{3}{2}$.



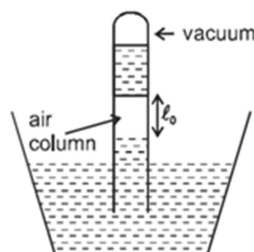
Both piston can slide freely in the chamber and do not allow any leakage of gas from one part of another. Gas in the left most part can be heated slowly using electric heater. At this situation force in rod is zero. Now heating assembly is switched on so that gas in left most part is heated slowly and the process continues till volume of right most part becomes $\frac{V_0}{8}$ and then heating assembly is switched off. Answer the following two questions for the process.

19. State equation for the thermodynamic process (assume the process to be polytropic of the form $PV^N = \text{constant}$) in the rightmost part of container is/are:
- (A) $PV^{4/3} = \text{constant}$ (B) $TV^{1/3} = \text{constant}$ (C) $PV^{5/3} = \text{constant}$ (D) $TV^{2/3} = \text{constant}$

20. Choose the correct option(s). -
 (A) Final temperature in middle part of container is $2 T_0$.
 (B) Final pressure in rightmost part of container is $16 P_0$.
 (C) Heat supplied by the heater during the process is $32 P_0 V_0$.
 (D) Magnitude of heat flow from conducting piston is $2 P_0 V_0$.
21. One side of vertical U-tube contains water and other contains oil as shown in figure. The ends of the tube are connected to the two vessels of same volume. Vessel A is filled with air at atmospheric pressure and vessel B is also filled with air but the absolute temperature inside vessel B is one fourth that of vessel A and number of moles inside vessel B is 5 times that of vessel A. The relative density of the oil is:
 (atmospheric pressure $= 10^5$ Pa, $g = 10 \text{ m/s}^2$)

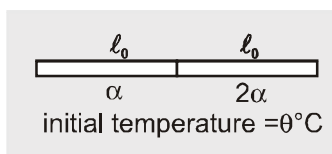


22. At the middle of the mercury barometer tube there is a little column of air with the length ℓ_0 and there is vacuum at the top as shown. Under the normal atmospheric pressure and the temperature of 300 kelvin, $\ell_0 = 10 \text{ cm}$. What will be the length(in cm) of the air column if the temperature rises to 330 kelvin? (Neglect expansion of the tube) -

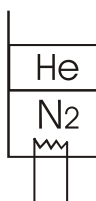


23. The temperature at the bottom of a pond of depth $L = 15 \text{ m}$ is 4°C . The temperature of the air, just above the layer of ice frozen at the pond's surface is -2°C for the past many days. The thermal conductivity of ice is three times that of water. Find the depth (in m) of unfrozen water below ice ?
24. The peak emission from a black body at a certain temperature occurs at a wavelength of $6000 \text{ \AA}$. On increasing its temperature, the total radiation emitted is increased 16 times. These radiation are allowed to fall on a metal surface. Photoelectrons emitted by the peak radiation at higher temperature can be brought to rest by applying a potential equivalent to the excitation potential corresponding to the transition for the level $n = 4$ to $n = 2$ in the Bohr's hydrogen atom. The work function of the metal is given by $\left(\frac{\alpha}{5}\right) \text{ eV}$ where α is the numerical constant. Find the value of α [Take : $hc = 12450 \text{ eV-\AA}$]:

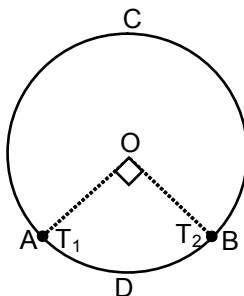
25. A gas consisting of rigid di-atomic molecules (degree of freedom = 5) at pressure $P_0 = 10^5 \text{ N/m}^2$ and temperature 273 K was compressed adiabatically to $\frac{1}{5}$ of its initial volume. The mean kinetic energy of rotation of molecules in final state is $n \times 10^{-21} \text{ J}$ (approx). Find value of 'n'. ($K = 1.38 \times 10^{-23} \text{ J/K}$, $(5)^{2/5} = 1.90$).
26. Two uniform rods of equal lengths (ℓ_0) and equal masses have coefficient of linear expansion α and 2α are placed in contact on a smooth horizontal surface as shown. The temperature of system is $\theta^\circ\text{C}$. Now the temperature is increased by $\Delta\theta^\circ\text{C}$. The junction of the rods will shift from its initial position by $\frac{\ell_0\alpha\Delta\theta}{n}$ where n is :



27. 5 moles of Nitrogen gas are enclosed in an adiabatic cylindrical vessel. The piston itself is a rigid light cylindrical container containing 3 moles of Helium gas. There is a heater which gives out a power 100 cal/second to the nitrogen gas. A power of 30 cal/sec is transferred to Helium through the bottom surface of the piston. The rate of increment of temperature of the nitrogen gas is (assuming that the piston moves slowly) :

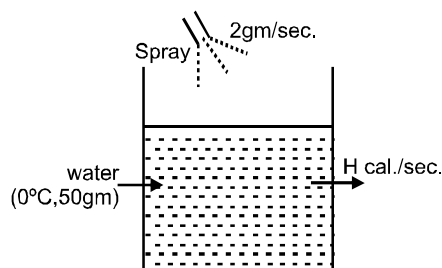


28. A ring consisting of two parts ADB and ACB of thermal conductivity K_1 carries heat at rate H from A to B. The ADB part is now replaced with another metal of (conductivity K_2) keeping same the temperature difference between A and B. The rate of heat carried increases to $2H$. If the ratio $\frac{K_2}{K_1} = \frac{n}{3}$. Find the value of n. (Given $\angle AOB = 90^\circ$) :

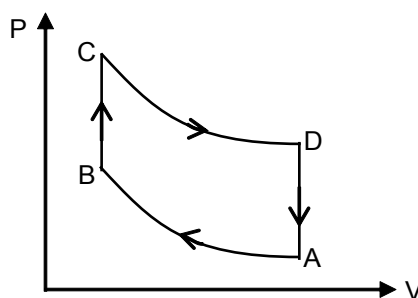


29. Two taps fill a tank with different flow rates. Tap A fills the tank by liquid L_1 in one hour and tap B fills the tank by liquid L_2 in 3 hours. Densities of two liquids are 1 g/cm^3 and 4 g/cm^3 respectively. Two taps throws liquids at temperature 10°C and 50°C respectively. If we open two taps together to fill the tank the final temperature of water is 20°C . Find the ratio of specific heat of liquid L_1 and liquid L_2 .

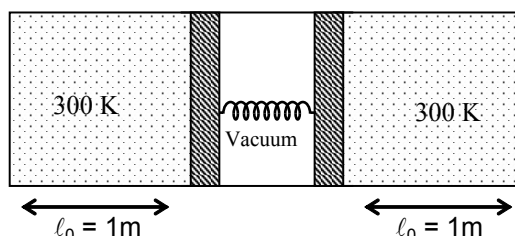
30. A container (water equivalent of container is negligible) contains 50 gm of water at 0°C . If H amount of heat is extracted per second from water and keeping the spray closed, then whole water freezes in 10 sec. If spray is also working and sprays the water with 2 gm/sec at 0°C , then time delay is $\frac{5x}{3}$ seconds in complete freezing compared to previous case keeping the rate of extraction of heat from water same. Then calculate 'x'. (Latent heat capacity of ice = 80 cal/gm)



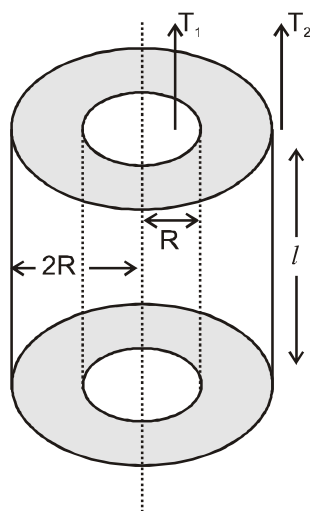
31. One mole of oxygen undergoes a cyclic process in which volume of the gas changes by a factor of 10 within the cycle as shown in figure. The processes AB and CD are adiabatic while process BC and DA are isochoric. What is the efficiency of the process in % ? ($10^{0.4} \approx 2.5$)



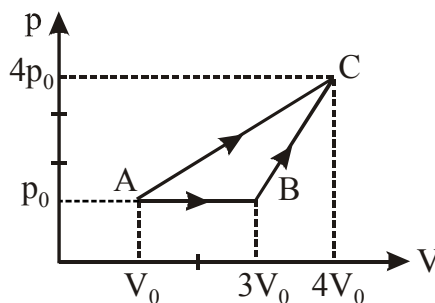
32. Consider the shown diagram where the two chambers separated by piston-spring arrangement contain equal amounts of same ideal gas. Initially when the temperatures of the gas in both the chambers are kept at 300 K. The compression in the spring is 10 cm. The temperature of the left and the right chambers are now raised slowly to 400 K and 500 K respectively. The pistons are free to slide (i.e. no friction between piston and chamber). Final compression in the spring is found to be $\frac{\alpha}{60}$ m. Find α . (Use $\sqrt{481} \approx 22$)



33. Inner surface of a cylindrical shell of length ℓ and of material of thermal conductivity k is kept at constant temperature T_1 and outer surface of the cylinder is kept at constant temperature T_2 such that $(T_1 > T_2)$ as shown in figure. Heat flows from inner surface to outer surface radially outward. Inner and outer radii of the shell are R and $2R$ respectively. Due to lack of space this cylinder has to be replaced by a smaller cylinder of length $\frac{\ell}{2}$ inner and outer radii $\frac{R}{4}$ and R respectively and thermal conductivity of material nk . If rate of radially outward heat flow remains same for same temperatures of inner and outer surface i.e. T_1 and T_2 , then find the value of n .

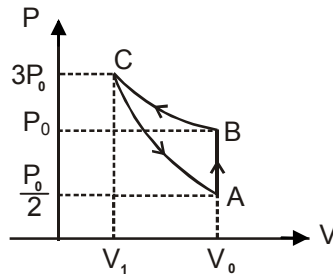


34. A certain quantity of ideal gas takes up 56 J of heat in the process AB and 360 J in the process AC. What is the number of degrees of freedom of the gas.

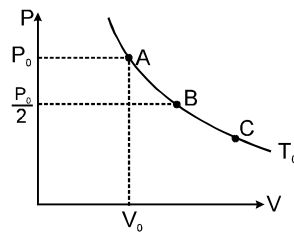


35. N molecules each of mass m of gas A and $2N$ molecules each of mass $2m$ of gas B are contained in a vessel which is maintained at a temperature T . The mean square of velocity of the molecules of B type is denoted by V^2 and the mean square of the x-component of the velocity of A type is denoted by w^2 . If the ratio of $\frac{w^2}{V^2}$ is $\frac{n}{3}$, find n ?
36. Let the wavelength at which the spectral emissive power of a black body (at a temperature T) is maximum, be denoted by $\lambda_{\max}$. As the temperature of the body is increased by 1 K, $\lambda_{\max}$ decreases by 1 percent. The temperature T (in K) of the black body is:
37. A monoatomic gas undergoes a process given by $2 dU + 3 dW = 0$, then the process is PV^N , where N is :

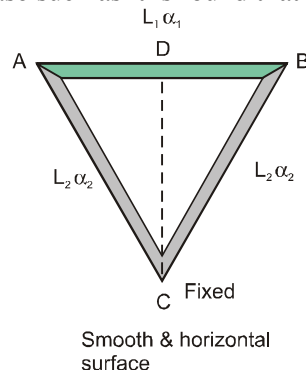
38. One mole of an ideal gas undergoes a thermodynamic cyclic process as shown in the figure. The cyclic process consists of an isochoric, an isothermal and an adiabatic process. Adiabatic exponent of gas is $\frac{\ln(N)}{\ln 3}$, where N is:



39. An insulated chamber at a height h above the earth's surface and maintained at 30°C has a clock fitted with an uncompensated pendulum. The maker of the clock for the chamber mistakenly designs it to maintain correct time at 20°C at that height. It is found that if the chamber were brought to earth's surface the clock in it would click correct time at 30°C . The coefficient of linear expansion of the material of pendulum is $\frac{h}{NR}$, where N is: (earth's radius is R)
40. The state of an ideal gas is changed through an isothermal process at temperature T_0 as shown in figure. The work done by gas in going from state B to C is double the work done by gas in going from state A to B. If the pressure in the state B is $\frac{P_0}{2}$ then the pressure of the gas in state C is $\frac{P_0}{N}$ where N is:



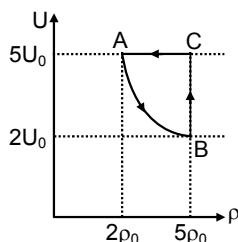
41. Figure shows isosceles triangle frame ABC of two different material shown in figure. Thermal expansion coefficient of the rod ADB is α_1 and for rod ACB is α_2 . End C is fixed and whole system is placed on smooth horizontal surface and D is midpoint of rod AB and CD is perpendicular to the AB. If temperature of the system is increase such as it is found that distance CD remain fixed then.



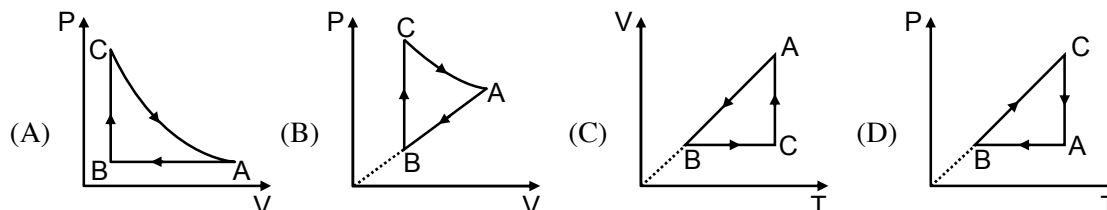
- (A) $\frac{L_1}{L_2} = 2\sqrt{\frac{\alpha_2}{\alpha_1}}$ (B) $\frac{L_1}{L_2} = 2\sqrt{\frac{\alpha_1}{\alpha_2}}$ (C) $\frac{L_1}{L_2} = \sqrt{\frac{\alpha_1}{\alpha_2}}$ (D) $\frac{L_1}{L_2} = \sqrt{\frac{\alpha_2}{\alpha_1}}$

Paragraph for Question Nos. 42 & 43

Figure shows variation of internal energy U with the density ρ of one mole of ideal mono-atomic gas for a thermodynamics cycle ABCA. Here process AB is a part of rectangular hyperbola. Answer the following two questions.



42. Incorrect diagram for the above process is:



43. Which of the following options is incorrect:

- (A) Work done by gas in process AB is $-2U_0$
 (B) Work done by gas in process BC is zero
 (C) Net work done by the gas during one cycle is $\left(\frac{10}{3} \ln 2.5 - 2\right)U_0$
 (D) Heat supplied to the gas during process CA is $\left(\frac{10}{3} \ln 2.5 - 1\right)U_0$

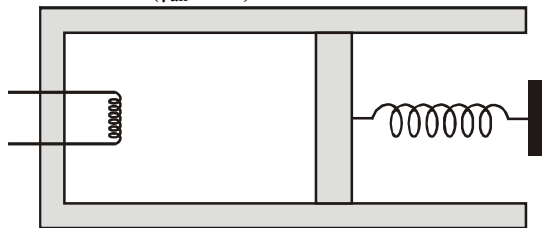
Paragraph for Question Nos. 44 to 46

Temperature inside the thin hollow sphere (which is placed in vacuum) of surface area 1 m^2 is maintained at 100°C with the help of an electric heater of constant power. The outer surface of sphere emits 120 W power in radiation and absorbs 50 watt radiation incident on it at steady state. It is also observed in other experiment with same sphere that it loses heat at a rate 50 watt power when temperature of outer surface is 40°C . (Assume temperature of the surrounding is constant (20°C) and Newtons law of cooling is valid, thermal conductivity of material is $7 \times 10^{-4} \text{ Watt/m } ^\circ\text{C}$)

44. Electric power of heater is :
 (A) 50 watt (B) 60 watt (C) 20 watt (D) 70 watt
45. Temperature of outer surface of sphere is :
 (A) 100°C (B) 80°C (C) 48°C (D) 60°C
46. Thickness of hollow sphere is :
 (A) 0.6 mm (B) 0.52 mm (C) 0.7 mm (D) none of these

Paragraph for Question Nos. 47 to 48

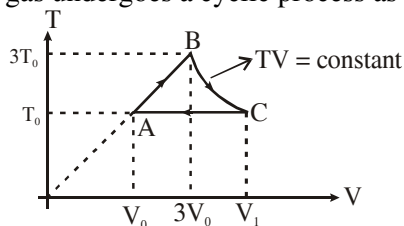
A piston encloses some air in the cylindrical vessel with horizontal longitudinal axis as shown in the drawing. Initial pressure of the air is equal to the external atmospheric pressure of 10^5 Pa. The cross-sectional area of the piston is 300 cm^2 . An originally unstretched spring with spring constant 2000 N/m is attached the piston. The walls of the vessel and the piston are perfectly insulated. The initial volume of the enclosed air is 0.024 m^3 . Its initial temperature is 300 K . The air is heated to 360 K with a heating filament built into the vessel. ($\gamma_{\text{air}}=1.4$)



47. Find the displacement of the piston caused by the heating :
 (A) 0.1 m (B) 0.2 m (C) 0.3 m (D) 0.4 m
48. Change in internal energy of the gas is :
 (A) 800 J (B) 1200 J (C) 2000 J (D) 200 J

Paragraph for Question No. 49 & 50

One mole of an ideal diatomic gas undergoes a cyclic process as shown in figure.



49. Heat capacity of process $B \rightarrow C$ is :
 (A) $\frac{R}{2}$ (B) $\frac{3R}{2}$ (C) $\frac{5R}{2}$ (D) $2R$
50. The efficiency of cycle is :
 (A) $\frac{4 - 2\ln 3}{7}$ (B) $\frac{4 + 2\ln 3}{7}$ (C) $\frac{4 - \ln 3}{7}$ (D) $\frac{2 - \ln 3}{7}$

ANSWER KEY OF DPP NO. 07 (JEE-ADVANCED)

1.	(A,C)	2.	(A,B,C,D)	3.	(A,B)	4.	(A,C)	5.	(B,D)
6.	(A,C)	7.	(A,B)	8.	(A,C,D)	9.	(A,B)	10.	(B,C)
11.	(A,B)	12.	(A,C,D)	13.	(A,D)	14.	(A,C)	15.	(A,C)
16.	(A,C)	17.	(A,B,D)	18.	(A,B,C)	19.	(A,B)	20.	(A,B,C,D)
21.	0.75	22.	11.00	23.	06.00	24.	08.00	25.	07.00
26.	04.00	27.	02.00	28.	07.00	29.	04.00	30.	04.00
31.	60.00	32.	09.00	33.	04.00	34.	05.00	35.	02.00
36.	100.00	37.	02.00	38.	06.00	39.	05.00	40.	08.00
41.	(A)	42.	(B)	43.	(D)	44.	(D)	45.	(C)
46.	(B)	47.	(A)	48.	(B)	49.	(B)	50.	(A)